CHAPTERS 24 TO 31 · COMPLETE GUIDE

Business AI Stack Handbook

A practical guide with steps, examples and a checklist. Read the chapter, then build the example.

CHAPTER 24 · BUSINESS AI STACK

The business AI stack: which tool does what, and where to learn it

Twelve tools cover most small-business automation. Each row below names the tool, the job it does for you, and where SkillNavora teaches it.

Category	Recommended tool	Your purpose	Where you learn it
Automation	n8n	Main workflow engine	Automation module (Module 20) and the n8n lessons in the agents track
Alternative automation	Make	Faster visual prototyping	Automation module (Module 20) and the Make tutorial
AI models	OpenAI API	Classification, extraction, reasoning	GenAI lessons and the AI Tools Lab
Alternative AI	Anthropic Claude API	AI workflows and document processing	Claude for daily PM work (Module 14) and Claude Code (Module 15)
Database	Supabase	Customer and workflow data	Agents and GenAI lessons; the SkillNavora LMS runs on Supabase
Backend	Python	Custom business logic	Python track in the Data Science & AI handbook
Frontend	Next.js	Customer-facing dashboards	**New: Chapter 25**
CRM	Zoho CRM	Customer management	**New: Chapter 26**
Communication	WhatsApp Business Platform	Customer messaging	**New: Chapter 27**
Analytics	Power BI	Business reporting	Power BI track in the Data Science & AI handbook
Deployment	Vercel and AWS	Application hosting	**New: Chapter 28**
Project management	Jira	Product and development tracking	**New: Chapter 29**

How the tools fit together

A typical flow: a lead arrives on your landing page (Chapter 31), it is saved in your database (Supabase), an automation (n8n or Make) adds it to your CRM (Zoho), sends a WhatsApp message (Chapter 27), and an AI model (OpenAI or Claude) reads any free-text message and sorts it. Reports go to Power BI, and the product team tracks the work in Jira.

START SMALL

Pick one flow, such as "website enquiry to WhatsApp reply". Build it end to end with two or three tools before adding the next. A working small flow beats a large plan.

Checklist before you buy any tool

- What decision or task does it replace?
- Who will use it each day, and do they need training?
- Where does the customer's data live, and can you export it?
- What does it cost at 10 times your current volume? Check each vendor's current pricing page.

Official documentation for the tools in this chapter: n8n, OpenAI API, Supabase, WhatsApp developer platform and Python. Check these first when an API detail or version has changed.

CHAPTER 25 · BUSINESS AI STACK

Next.js: customer-facing dashboards

Next.js is a React framework for building websites and web apps. Use it when a customer or your team needs a dashboard that logs in, shows live data and works on a phone.

When to use Next.js

- A dashboard that customers log into to see their orders, enquiries or reports.
- A public site that needs fast pages, search engine ranking and server-side data.
- A tool your team uses that reads data from Supabase or an API.

The core ideas, in plain words

- **Pages and routes.** Each file in the `app/` folder becomes a page at the matching address. `app/dashboard/page.js` becomes `/dashboard`.
- **Server and client components.** Components run on the server by default, which keeps secrets off the browser. Add `"use client"` only where you need clicks, forms or live updates.
- **Data fetching.** Read data on the server, then pass it to the page. Never put a secret key in client code.
- **Environment variables.** Store keys in `.env.local` and never commit that file to GitHub.

Build a simple enquiry dashboard

1. Create the project: `npx create-next-app@latest enquiry-dashboard` and accept the defaults.
2. Create a Supabase project and a table `enquiries` with the columns `name`, `phone`, `message`, `status` and `created_at`.
3. Add the Supabase URL and the publishable key to `.env.local`. Use the publishable key only; never use the service-role key in the browser.
4. In `app/dashboard/page.js`, read the latest 50 enquiries on the server and render them as a table with status labels.
5. Protect the page: only show it after sign-in, and make sure the database rules (row level security) allow each user to see only their own rows.

```
// app/dashboard/page.js (server component, simplified)
import { createClient } from "@supabase/supabase-js";
const supabase = createClient(process.env.NEXT_PUBLIC_SUPABASE_URL,
process.env.NEXT_PUBLIC_SUPABASE_KEY);
export default async function Dashboard() {
  const { data } = await supabase.from("enquiries").select("*").order("created_at", { ascending: false
}).limit(50);
  return <table>{data.map(e => <tr key={e.id}><td>{e.name}</td><td>{e.status}</td></tr>)}</table>;
}
```

COMMON MISTAKES

Putting a secret key in a client component. Forgetting row level security, so every user sees every row. Shipping a dashboard with no sign-in.

Checklist

- Pages load without errors on a phone.
- No secret keys in the code or the browser.
- Only signed-in users see the dashboard.
- The project is in GitHub, ready for Chapter 28 (deployment).

Reference: the Next.js documentation (nextjs.org/docs) and the Supabase documentation for the client library.

CHAPTER 26 · BUSINESS AI STACK

Zoho CRM: managing customers and leads

A CRM (customer relationship management) system keeps every lead, customer and conversation in one place, with a clear next step for each one.

What Zoho CRM does for a small business

- **Leads and contacts.** Every enquiry becomes a lead, which becomes a contact once you talk to them.
- **Deals (pipeline).** Each sales opportunity moves through stages, such as New, Contacted, Demo, Proposal and Won or Lost.
- **Tasks and follow-ups.** A reminder for each lead, so nobody is forgotten.
- **Reports.** How many leads came in, which source they came from, and how many became customers.

Set up a clean pipeline

1. Create a sign-up account and set your company details, time zone and currency.
2. Define the stages. Keep them to five or six, each with a clear exit rule (for example, "Demo booked" means a date is in the calendar).
3. Add the lead fields you really use: source, interest, budget range, preferred time. Avoid fields nobody fills in.
4. Create lead-assignment rules, so each new lead goes to one person with a response deadline.
5. Connect your website form and WhatsApp (Chapter 27) with an automation tool such as n8n or Make, so leads arrive without copy-paste.

Connecting Zoho to the rest of the stack

Zoho provides its own APIs and connectors. A common pattern: the website form saves the lead to Supabase, an n8n workflow creates the lead in Zoho through the Zoho API, and the same workflow sends a WhatsApp confirmation. Check the current Zoho API documentation for the exact endpoints and the authentication method before you build.

DATA HYGIENE

Remove duplicates once a month, record consent for marketing messages, and give customers a way to opt out. These protect you and keep the pipeline honest.

Metrics to watch

- Lead response time (target: within one working hour).
- Conversion from each stage to the next.
- Lead source that brings the most paying customers, not just the most leads.

Reference: Zoho CRM help and developer documentation. Plan limits and prices change; check the current plan page.

CHAPTER 27 · BUSINESS AI STACK

WhatsApp Business Platform: customer messaging

WhatsApp is where many customers already talk. The WhatsApp Business Platform lets your systems send and receive messages automatically, with your own number and business name.

Key concepts

- **WhatsApp Business account and phone number.** The number your customers message. It is registered through Meta Business Suite.
- **Customer service window.** After a customer messages you, you can reply with free-form text for 24 hours. After that, you may only send approved template messages.
- **Template messages.** Pre-approved messages with placeholders, for example "Hi {{1}}, your demo is on {{2}}". Meta reviews each template before it can be used.
- **Webhooks.** Meta sends each incoming message to a URL you control. Your automation reads it and replies.

A lead-reply flow

1. A visitor fills the enquiry form on your site. The lead is saved in Supabase.
2. An n8n or Make workflow sends an approved template: "Thanks {{1}}, a counsellor will call you within one working hour."
3. If the customer replies, the webhook receives the message. An AI model (OpenAI or Claude) classifies the intent: price, schedule, complaint, or other.
4. Simple questions get an automatic answer inside the 24-hour window. Anything else goes to a person in the CRM (Chapter 26).

```
// Sending a text reply (Cloud API shape; check the current version in the WhatsApp docs)
POST https://graph.facebook.com/{API_VERSION}/{PHONE_NUMBER_ID}/messages
Authorization: Bearer {ACCESS_TOKEN}
{ "messaging_product": "whatsapp", "to": "9198XXXXXXXX", "type": "text", "text": { "body": "Thanks, we
will call you." } }
```

RULES THAT KEEP YOU OUT OF TROUBLE

Only message people who opted in. Honour "STOP" immediately. Do not send marketing content as a utility template. Keep a record of consent with the date and the source.

Checklist

- Business verified and display name approved.
- Templates approved before launch.
- Webhook verification passes, and each message is saved.
- A human handover path for anything the AI is unsure about.

Reference: the WhatsApp developer documentation (Cloud API), and the Meta Business Suite help pages for account setup. Message limits and pricing change; check the current pricing page before you launch.

CHAPTER 28 · BUSINESS AI STACK

Deployment with Vercel and AWS

Deployment means putting your app on a server so customers can use it. Vercel is the simplest option for Next.js apps. AWS gives you full control for systems that need it.

Which to choose

Need	Vercel	AWS
Next.js website or dashboard	Easiest: connect GitHub, every push deploys	Possible, more setup (for example Amplify or containers)
Background jobs, custom servers, private networks	Limited	Strong: EC2, Lambda, ECS, RDS and more
Who should run it	A small team with no DevOps	A team that owns its infrastructure

Deploy a Next.js app to Vercel

1. Push your project to a GitHub repository. Keep `.env.local` out of Git (check `.gitignore`).
2. In Vercel, choose "Add new project", pick the repository and accept the detected Next.js settings.
3. Add the environment variables (Supabase URL, publishable key, any API keys) in the project settings for Production and Preview.
4. Deploy. Each push to the main branch creates a new production build; each branch gets a preview link.

When you move to AWS

- Use separate accounts or roles for development and production.
- Store secrets in AWS Secrets Manager or Parameter Store, never in code.
- Turn on billing alerts before you launch.
- Use the least-privilege principle: each service gets only the permissions it needs.

SECURITY BASICS FOR ANY HOST

Use HTTPS only. Keep keys in environment variables. Rotate any key that was ever pasted into chat or a public repository. Limit who can change production settings.

Checklist

- Production URL loads over HTTPS.
- Environment variables are set for production.
- A rollback plan: you can redeploy the previous version in one click (Vercel) or from the last tag (AWS).

Reference: Vercel documentation for Next.js deployments and the AWS documentation for the services you choose. Pricing depends on usage; check the current pricing pages.

CHAPTER 29 · BUSINESS AI STACK

Jira: tracking product and development work

Jira is a work-tracking tool used by product and engineering teams. It keeps the backlog, the sprint plan and the status of each item in one place.

The Jira hierarchy

- **Epic.** A large goal, for example "Customer enquiry dashboard".
- **Story.** A user-facing piece of work: "As a counsellor, I can see new enquiries first".
- **Task and bug.** The technical steps and defects behind a story.
- **Sprint.** A time-boxed set of stories, usually two weeks.

Set up a project in 30 minutes

1. Create a Scrum or Kanban project. Scrum suits fixed sprints; Kanban suits continuous flow.
2. Set the workflow: To do, In progress, In review, Done. Keep it to four or five statuses.
3. Write each story with a clear acceptance criterion: "Given a new enquiry, when it is saved, then the counsellor sees it within one minute."
4. Estimate in story points, plan one sprint, and review the board every day in a 15-minute stand-up.

Connecting Jira to the rest of the stack

Link commits and pull requests from GitHub to the Jira issue key (for example `SN-42`), so every change is traceable. Automation rules can move an issue when its pull request is merged.

> **GOOD HABIT**
> Keep the backlog short and prioritised. A backlog of 300 untriaged items is not a plan; it is a list of things nobody will do.

Checklist

- Every story has an owner and acceptance criteria.
- The sprint goal is one sentence.
- Done means done: tested, reviewed and deployed.

Reference: Atlassian's Jira documentation for project setup, workflows and automation rules.

CHAPTER 30 · DIGITAL MARKETING

Meta Business Suite, social media marketing and Google Ads (PPC)

Paid and organic channels work differently. This chapter explains how each one works, what it costs you in time or money, and how to set up a first campaign safely.

1. Meta Business Suite: one place for Facebook and Instagram

- **Business portfolio.** The top-level account that owns your pages, ad accounts and people. Give team members the least access they need.
- **Pages and Instagram accounts.** Your public presence. Connect the Instagram account to the business portfolio.
- **Ad accounts and Meta Ads Manager.** Where paid campaigns are built and paid for.
- **Inbox.** Messages from Facebook, Instagram and WhatsApp in one place.

2. Social media marketing: organic growth

Organic posts reach followers and, sometimes, more people when they are shared. Growth comes from consistency and useful content: short videos that answer one question, carousels that teach a method, and case studies with real numbers. Track reach, saves, shares and profile visits, not only likes.

Format	Good for	Example
Short video (reel)	Reach to new people	"3 mistakes in a sales follow-up"
Carousel	Teaching a method	"A 5-step lead qualification checklist"
Case study post	Trust and enquiries	"How one clinic doubled bookings in 6 weeks"

3. Paid social on Meta: how it works

1. **Objective.** Choose the goal the platform optimises for: awareness, traffic, leads or sales. The choice changes who sees the ad.
2. **Audience.** Location, age, interests and custom or lookalike audiences built from your customer list.
3. **Budget and bidding.** A daily or lifetime budget. The auction chooses which ad to show, based on bid, predicted response and relevance.
4. **Creative.** The image or video and the text. Creative is usually the biggest lever on results.

4. Google Ads (PPC): how it works

Pay-per-click (PPC) ads appear in Google Search when someone types a keyword you bid on. You pay only when someone clicks. The ad auction weighs your bid, the expected click-through rate, the quality of your landing page and the ad's relevance.

- **Keywords and match types.** Choose intent-rich phrases such as "PM course online" and use negative keywords to block waste, such as "free" if you do not offer a free course.
- **Quality score.** Google rates each keyword's relevance, expected clicks and landing-page experience. A higher score lowers your cost per click.
- **Conversion tracking.** Set up a conversion for each enquiry or purchase. Without it you are buying clicks, not customers.

START WITH A SMALL TEST

Begin with a small daily budget on 3 to 5 keywords and one landing page. Run for at least two weeks before judging. Stop any keyword with clicks and no conversions after a reasonable spend.

5. Measure what matters

- Cost per lead and cost per customer, not cost per click.
- Lead quality: how many became paying customers.
- Payback period: how long the marketing spend takes to earn back.

Checklist

- Business portfolio and access roles are set up.
- Conversion tracking is tested with a real enquiry.
- Each campaign has one goal, one audience and one landing page.
- Weekly review: pause, test or scale.

Reference: Meta Business Help Centre and Google Ads Help. Ad policies, bidding options and interface names change; check the current help pages before each campaign.

CHAPTER 31 · DIGITAL MARKETING

A one-page website for any firm that generates leads

A one-page site is one clear offer, one proof section and one way to act. It works for a clinic, a coaching business, a local service or a B2B firm.

The structure that converts

Section	What to write
1. Headline	The result for the customer, in one line: "More booked appointments in 30 days, without extra ads."
2. Proof	A number, a logo the customer allowed you to use, or a short testimonial with a name and role.
3. The offer	What is included, how long it takes, and the price range or a clear next step.
4. How it works	Three steps: contact, plan, result.
5. Objections	Three questions with honest answers: cost, time, what if it does not work.
6. Form and call to action	A short form (name, phone, need) and a WhatsApp button. One primary action only.

Build it in a day

1. Write the offer on paper first. If you cannot say it in two sentences, the page will not work.
2. Build the page with a no-code tool or Next.js (Chapter 25). Keep it to one screen on a phone, with the call to action visible without scrolling.
3. Connect the form to Supabase or your CRM (Chapter 26), and send an instant WhatsApp reply (Chapter 27).
4. Add tracking: a page view, a form submit and a WhatsApp click. Without these you cannot tell what works.
5. Send traffic: a Google Ads search campaign (Chapter 30) to a small list of keywords, and a Meta post to people who already engaged.

Turning enquiries into customers

- Reply within one working hour. Speed is the biggest single factor in converting leads.
- Qualify with two questions: what is the need, and when do you want to start.
- Follow up three times over a week, then stop. Record every outcome in the CRM.

LEAD-GENERATION FUNNEL TO MEASURE

Visitors → form submits (conversion rate) → replies within an hour → qualified leads → customers. Track each step weekly; the weakest step is where to work next.

Checklist

- One offer, one call to action.
- Works on a phone at 390 pixels wide.
- Form, WhatsApp and CRM tested with a real submission.
- Privacy notice and consent checkbox on the form.

Reference: the documentation of the tools you use (for example Supabase for form storage and the WhatsApp developer platform for messages).